

Opinion Pooling IV: Promoting Group Epistemic Rationality

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Accuracy and Justification in Pooling

We've examined three families of pooling strategies:

- **Linear pooling:** Weighted averages (satisfies most axioms, but violates External Bayesianity)
- **Geometric pooling:** Geometric means over partitions (satisfies External Bayesianity, but partition-sensitive)
- **Multiplicative pooling:** Generalization of geometric pooling (no unanimity preservation)

Today: Moving beyond axioms to evaluate pooling by its **epistemic consequences**

Two Key Questions

How well do pooling strategies promote:

1. **Accuracy:** Getting closer to the truth
2. **Justification:** Preserving epistemic warrant

These relate to our earlier discussion of Group Rationality: groups should have accurate, justified beliefs.

Measuring Accuracy

Core idea: A credence can't be true or false, but it can be more or less *accurate*.

- If I'm 60% confident it will rain and you're 5% confident
- If it rains: my credence was more accurate
- If it doesn't rain: your credence was more accurate

Can we measure accuracy of *entire credence functions*, not just single credences?

The Ideal Credence Function

For any world w , the ideal credence function assigns:

- Credence **1** to all propositions true at w
- Credence **0** to all propositions false at w

This represents perfect knowledge of w .

Accuracy = how close your credences are to the ideal

Inaccuracy = how far your credences are from the ideal

Basic idea: Sum up the “mistakes” across all your credences.

Example: You’re 40% confident it will rain.

- If it rains (ideal: 100% confident), you’re off by 60 percentage points
- If it doesn’t rain (ideal: 0% confident), you’re off by 40 percentage points

Different ways to aggregate these errors give different accuracy measures.

If you use appropriate accuracy measures (technical details omitted):

- **Probabilism vindicated:** If your credences violate probability rules, there's always a better credence function that's guaranteed to be more accurate
- **Bayesianism vindicated:** Updating by Bayes' Rule minimizes expected inaccuracy

This provides justification for fundamental epistemic norms!

Accuracy-Based Arguments for Pooling

A huge community of Shakespeare scholars has credences about who authored the works attributed to Shakespeare.

- Christopher Marlowe?
- Francis Bacon?
- Edward de Vere?
- Shakespeare himself?

We want to summarize their collective view using a pooling strategy.

Suppose we pool to get group credence Q .

Problem: What if there's an alternative Q^* such that *every individual scholar* thinks Q^* would be more accurate than Q (in expectation)?

Then we've ascribed the wrong credences to the group — everyone agrees there's a better option!

Accuracy Consensus: There should be no alternative group credence that all individuals prefer (in terms of expected accuracy).

This is like a Pareto optimality condition for credence functions.

The result: Linear pooling is the *only* strategy that satisfies Accuracy Consensus.

(More precisely: if Q is not a linear pool, there exists an alternative that everyone prefers. If Q is a linear pool, no such alternative exists.)

Accuracy Consensus is important when individuals must “get behind” the group credence:

- Representing group opinion to outsiders (the Shakespeare authorship debate)
- Assessing group liability for collective actions (corporate responsibility cases)
- Not so important for individual peer disagreement responses

If accuracy is our goal for group credences, this favors linear pooling.

A search committee of 5 professors must decide on hiring a candidate.

- Prof. A (20 years experience): 80% confident the candidate will succeed
- Prof. B (15 years): 75% confident
- Prof. C (10 years): 60% confident
- Prof. D (5 years): 70% confident
- Prof. E (2 years): 85% confident

Questions for discussion:

1. Should they use equal weights or weight by seniority/expertise?
2. How would you justify your choice to the committee members?

Does accuracy favor a particular way of pooling imprecise credences?

Current status: Open question!

- No satisfactory theory of credal accuracy for imprecise credences yet
- Impossibility results: no measure has strict propriety for imprecise credences
- Work ongoing (Konek, 2019)

Interesting implication of Theorem 7:

For precise credences $P_1, \dots, P_n$, the convex hull $\text{conv}\{P_i : i = 1, \dots, n\}$ contains exactly those credence functions that are *not dominated* in expected accuracy.

This is precisely what the **IP pooling function** returns!

Suggests an accuracy-based argument for IP pooling (at minimum, rules out functions outside the convex hull).

The Diversity Prediction Theorem

Pooling strategies can aggregate any numerical estimates, not just credences.

County fair example: Lila (100 kg), Mona (94 kg), Nomy (93 kg) estimate the prize marrow's weight.

True weight: 91 kg.

The **average inaccuracy** of individual estimates exceeds the **inaccuracy of the straight linear pool**:

$$\frac{1}{3}(100-91)^2 + \frac{1}{3}(94-91)^2 + \frac{1}{3}(93-91)^2 > \left(\frac{100 + 94 + 93}{3} - 91 \right)^2$$

This is no coincidence!

Diversity Prediction Theorem: For any sequence of quantities with true values $T = (t_1, \dots, t_m)$ and individual estimates $A_i = (a_{i1}, \dots, a_{im})$:

$$\text{SED} \left(T, \frac{1}{n} \sum_{i=1}^n A_i \right) = \frac{1}{n} \sum_{i=1}^n \text{SED}(T, A_i) - \frac{1}{n} \sum_{i=1}^n \text{SED} \left(A_i, \frac{1}{n} \sum_{i=1}^n A_i \right)$$

Key insight: The second term is always positive when estimates differ (it measures diversity).

If estimates are not all identical ($A_i \neq A_j$ for some i, j), then:

$$\text{SED} \left(T, \frac{1}{n} \sum_{i=1}^n A_i \right) < \frac{1}{n} \sum_{i=1}^n \text{SED}(T, A_i)$$

Translation: The linear pool is more accurate than picking a random expert's estimate!

This holds *regardless* of:

- What quantities are being estimated
- What the true values are
- What estimates people give

Scenario: You're a local government official with no expertise on a housing project's consequences.

You consult multiple experts with different credences.

Options:

1. Pick one expert at random and defer to them
2. Pool the experts' credences and defer to the pool

The Diversity Prediction Theorem guarantees option 2 is more accurate in expectation!

But There's a Catch...

When deferring to experts, we want it to be *possible* that:

1. We think experts might disagree: $P(E_1 = E_2) < 1$
2. We defer to each expert individually: $P(X \mid E_1 = p) = p$ and $P(X \mid E_2 = q) = q$
3. We defer to the pool of experts:
$$P(X \mid E_1 = p, E_2 = q) = F(p, q)$$

All achieved by having appropriate unconditional and conditional credences.

Bad news: Linear pooling does *not* satisfy Deference Compatibility.

Researchers who've noticed this:

- Dawid, DeGroot & Mortera (1995)
- Bradley (2018)
- Gallow (2018)
- Dawid & Mortera (2020)
- Zhang (ms) — shows many other pooling functions also fail

Our local government official faces a trade-off:

- ☐ Linear pooling gives Diversity Prediction benefits
- ☐ Linear pooling violates Deference Compatibility

Which matters more depends on the context!

Maya works for a city government with no climate science background. She consults three climate scientists about flood risk from a proposed development:

- Dr. Chen: 70% chance of severe flooding in next 20 years
- Dr. Johnson: 45% chance
- Dr. Okafor: 60% chance

Questions for discussion:

1. Should Maya adopt the linear pool (58.3%) or choose one expert to defer to entirely?
2. What if she later learns Dr. Chen has relevant local expertise? How should she update?

The Principle of Coherent Minimal Mutilation

Inspiration: How do we combine conflicting information sources?

Scenario: Corporate liability case with executives Oskar, Prince, and Quentin.

We need to identify credences for the *executive team* that they cannot reasonably object to.

Executives might object if group credences diverge from individual credences *more than necessary*.

Idea: Group credence should minimize total “distance” from all individuals.

Question: But how do we measure “distance” between credence functions?

Approach 1: Sum of squared differences

- Like the Euclidean distance you learned in geometry
- Treats all dimensions equally

Approach 2: Information-theoretic distance

- Based on how “surprised” one distribution would be by another
- More sensitive to large discrepancies

Different distance measures give different pooling strategies:

- **Approach 1 (squared differences)** → Linear pooling!
- **Approach 2 (information-theoretic)** → Geometric pooling!

This helps explain why both strategies seem natural — they're each minimizing distance, just measured differently.

Both Accuracy Consensus and Minimal Mutilation have a **contractualist** character:

Group credences are legitimate when individuals have no grounds to complain:

- No complaint if no alternative is preferred by all (Accuracy Consensus)
- No complaint if no alternative has smaller total distance (Minimal Mutilation)

When to use: Cases where individuals must “get behind” the group credence (representation, liability).

A hospital review board must determine credences about whether a new surgical technique is safe:

- 3 surgeons have credences: 65%, 70%, 75%
- 2 anesthesiologists: 50%, 55%
- 1 nurse: 80%
- Their collective judgment will determine hospital policy

Questions for discussion:

1. Which pooling approach seems most fair to all board members?
2. Does it matter that their judgment creates policy that binds them all?
3. Should different professional roles be weighted differently?

Preserving Justification

Often we want pooling to preserve individual justification.

Idea: If each group member is justified in their credences, the group credences should also be justified.

But what does justification *mean* for group credences?

One natural condition (aligned with Lackey's GEAA view):

If individuals:

- Share the same prior credences
- Update on their own private evidence

Then pooling their posteriors should equal updating the shared prior on the *conjunction* of all individual evidence.

Evidence Pooling: For all probability functions and propositions $E_1, \dots, E_n$:

$$F(P_1(- \mid E_1), \dots, P_n(- \mid E_n))(X) = F(P_1, \dots, P_n)(X \mid E_1 \wedge \dots \wedge E_n)$$

This differs from External Bayesianity by allowing *different evidence* for each individual.

External Bayesianity is a special case (when all receive the same evidence).

Which Pooling Functions Satisfy This?

Good news: Geometric and multiplicative pooling over the finest partition (possible worlds $\mathcal{W}$) both satisfy Evidence Pooling!

(Bacelli & Stewart, 2023; Weisberg & Pettigrew, 2023)

But there's a problem...

To use geometric/multiplicative pooling over $\mathcal{W}$, we need each individual's credences over *every possible world*.

But: If we can collect that, we could just ask for their private evidence directly and pool that instead!

The pooling function becomes overkill for this purpose.

In science, we often have:

- Partition of **hypotheses**: $H_1, \dots, H_k$ (e.g., different theories)
- Partition of **evidence**: $E_1, \dots, E_m$ (e.g., experimental outcomes)
- These partitions don't nest within each other
- Each hypothesis assigns a *chance* to each piece of evidence

Examples:

- Physics: theories of particle behavior + experimental outcomes
- Climate science: different climate sensitivity values + radiosonde readings

Hypotheses $H_1, \dots, H_k$ specify different biases $q_1, \dots, q_k$ for a coin.

Evidence: N tosses with M heads.

By Bayes' Rule (with Principal Principle):

$$P(H_i | E) = \frac{P(H_i)q_i^M(1 - q_i)^{N-M}}{\sum_{j=1}^k P(H_j)q_j^M(1 - q_j)^{N-M}}$$

Taj has prior P_1 , witnesses N_1 tosses with M_1 heads (evidence E_1).

Ursula has prior P_2 , witnesses N_2 tosses with M_2 heads (evidence E_2).

Geometric pooling of their posteriors:

$$\begin{aligned} & \text{GP}_{\mathcal{H}}(P_1(- \mid E_1), P_2(- \mid E_2))(H_i) \\ &= \text{GP}_{\mathcal{H}}(P_1, P_2) \left(H_i \mid \frac{M_1 + M_2}{2} \text{ out of } \frac{N_1 + N_2}{2} \right) \end{aligned}$$

Problem: The sample size is *halved*! We update on half the evidence.

Multiplicative Pooling to the Rescue?

With multiplicative pooling (weights summing to $n = 2$):

$$\begin{aligned} & \text{MP}_{\mathcal{H}}(P_1(- \mid E_1), P_2(- \mid E_2))(H_i) \\ &= \text{MP}_{\mathcal{H}}(P_1, P_2)(H_i \mid (M_1 + M_2) \text{ out of } (N_1 + N_2)) \end{aligned}$$

Good news: Full sample size preserved!

Bad news: Multiplicative pooling doesn't satisfy Unanimity Preservation.

Even with shared priors, multiplicative pool of posteriors $\neq$ shared prior updated on total evidence.

This applies generally, not just to coin tosses:

For any hypotheses specifying chance functions where evidence pieces are probabilistically independent according to each chance function, the same pattern holds.

Discussion: The Archaeological Team

An archaeological team studies whether a site was inhabited in the Bronze Age:

- All three archaeologists started with the same prior (based on regional surveys): 40% confident
- Archaeologist 1 excavated area A, found 3 bronze artifacts out of 20 items
- Archaeologist 2 excavated area B, found 5 bronze artifacts out of 30 items
- Archaeologist 3 excavated area C, found 2 bronze artifacts out of 15 items

Questions for discussion:

1. Should they pool their posterior credences or pool all the evidence first?
2. Does the fact they physically separated to gather evidence matter?

Summary and Looking Ahead

Accuracy-based arguments:

- Accuracy Consensus characterizes linear pooling
- Diversity Prediction Theorem favors linear pooling for estimation
- But linear pooling violates Deference Compatibility
- Minimal Mutilation with SED gives linear pooling, with GKL gives geometric pooling

Justification-based arguments:

- Evidence Pooling: want pooled posteriors to equal pooled priors updated on total evidence
- Geometric/multiplicative pooling over $\mathcal{W}$ satisfy this, but impractical
- For realistic scientific cases with hypothesis/evidence partitions:
 - Geometric pooling halves sample size
 - Multiplicative pooling preserves sample size but loses unanimity

We now have three types of arguments for pooling strategies:

Approach	Favors
Axiomatic	Different strategies depending on which axioms we prioritize
Accuracy (Consensus)	Linear pooling
Accuracy (Diversity)	Linear pooling
Accuracy (Minimal Mutilation)	Depends on distance measure (SED $\rightarrow$ linear, GKL $\rightarrow$ geometric)
Justification	Geometric/multiplicative (with caveats)

No single pooling strategy satisfies all desiderata:

- Linear pooling: good for accuracy consensus and diversity, bad for External Bayesianity and deference compatibility
- Geometric pooling: good for External Bayesianity and justification, bad for partition sensitivity and sample size
- Context determines which trade-offs are acceptable

The expertise problem: How should we weight different individuals?

- Equal weights? (Democratic but potentially inaccurate)
- Expertise-based weights? (Accurate but who decides expertise?)
- This connects to social epistemology and democratic theory

Peer disagreement revisited: These accuracy arguments might inform *individual* responses to peer disagreement.

If accuracy matters, and if linear pooling promotes accuracy, perhaps individuals should adopt linear pools of their own credence and their peers' credences?

But: Deference Compatibility problem suggests caution here.

Next topics in opinion pooling:

- Pragmatic arguments: pooling for decision-making rather than representation
- Imprecise probability pooling: developing the theory further
- Applications to specific domains (scientific consensus, legal contexts, democratic deliberation)